

Homework 2

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Problem 1

We know that port 1 is matched, thus

$$Y_{AB} + Y_{AC} = \frac{1}{Z_{\infty 1}}$$

For the QWT:

$$Y_{AB} = \frac{Z_{r2}}{Z_{\infty 2}^2}, \quad Y_{AC} = \frac{Z_{r3}}{Z_{\infty 3}^2}$$

In this problem,

$$Z_{r1} = Z_{r2} = Z_{r3} = Z_0 = 50 \, \Omega.$$

Substituting,

$$\frac{50}{Z_{\infty 2}^2} + \frac{50}{Z_{\infty 3}^2} = \frac{1}{50}$$

From our course, we learn that

$$\frac{P_2}{P_3} = \frac{Z_{r2} Z_{\infty 3}^2}{Z_{r3} Z_{\infty 2}^2}$$

And in this problem

$$\frac{P_3}{P_2} = 1.25$$

Then we get

$$\frac{Z_{\infty 3}}{Z_{\infty 2}} = \sqrt{0.8}$$

Using this relation together with the matching equation, we get

$$Z_{\infty 2} = 75 \, \Omega, \quad Z_{\infty 3} \approx 67.1 \, \Omega$$

For $Z_{\infty 2}$:

$$\frac{W_2}{h} = 0.886$$

so

$$W_2 = 1.37 \text{ mm}$$

$$\varepsilon_{\text{eff}2} = 3.28$$

$$\lambda_{g2} = \frac{\lambda_0}{\sqrt{\varepsilon_{\text{eff}2}}} = 2.85 \text{ cm}$$

$$\ell_2 = \frac{\lambda_{g2}}{4} = 0.71 \text{ cm}$$

For $Z_{\infty 3}$:

$$\frac{W_3}{h} = 1.115$$

so

$$W_3 = 1.73 \text{ mm}$$

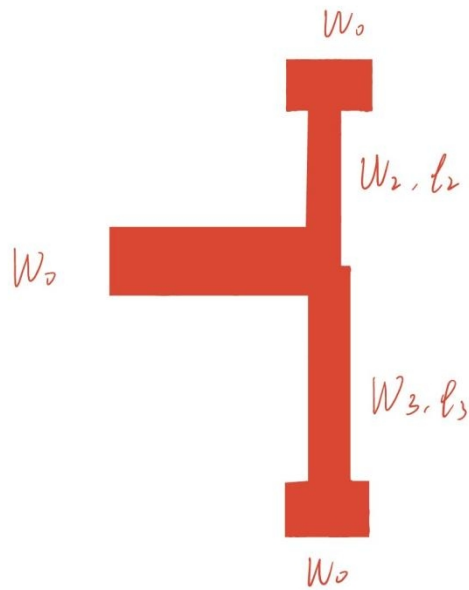
$$\lambda_{g3} = \frac{\lambda_0}{\sqrt{\varepsilon_{\text{eff}3}}} = 2.81 \text{ cm}$$

$$\ell_3 = \frac{\lambda_{g3}}{4} = 0.70 \text{ cm}$$

And

$$W_0 = 2.94 \text{ mm}$$

Thus we can draw the layout of the microstrip circuit:



As we learned in our course,

$$\Gamma_{L2} = \frac{Z_L - Z_{\infty 2}}{Z_L + Z_{\infty 2}}$$

$$\Gamma_{in,2}(f) = \Gamma_{L2} e^{-2jk_2(f)\ell_2}$$

with

$$k_2(f) = \frac{2\pi f \sqrt{\epsilon_{\text{eff}2}}}{c}$$

And

$$Z_{in,2}(f) = Z_{\infty 2} \frac{1 + \Gamma_{in,2}(f)}{1 - \Gamma_{in,2}(f)}$$

Similarly

$$\Gamma_{L3} = \frac{Z_L - Z_{\infty 3}}{Z_L + Z_{\infty 3}}$$

$$\Gamma_{in,3}(f) = \Gamma_{L3} e^{-2jk_3(f)\ell_3}$$

$$k_3(f) = \frac{2\pi f \sqrt{\epsilon_{\text{eff}3}}}{c}$$

$$Z_{in,3}(f) = Z_{\infty 3} \frac{1 + \Gamma_{in,3}(f)}{1 - \Gamma_{in,3}(f)}$$

The two branches are connected in parallel at point A, so

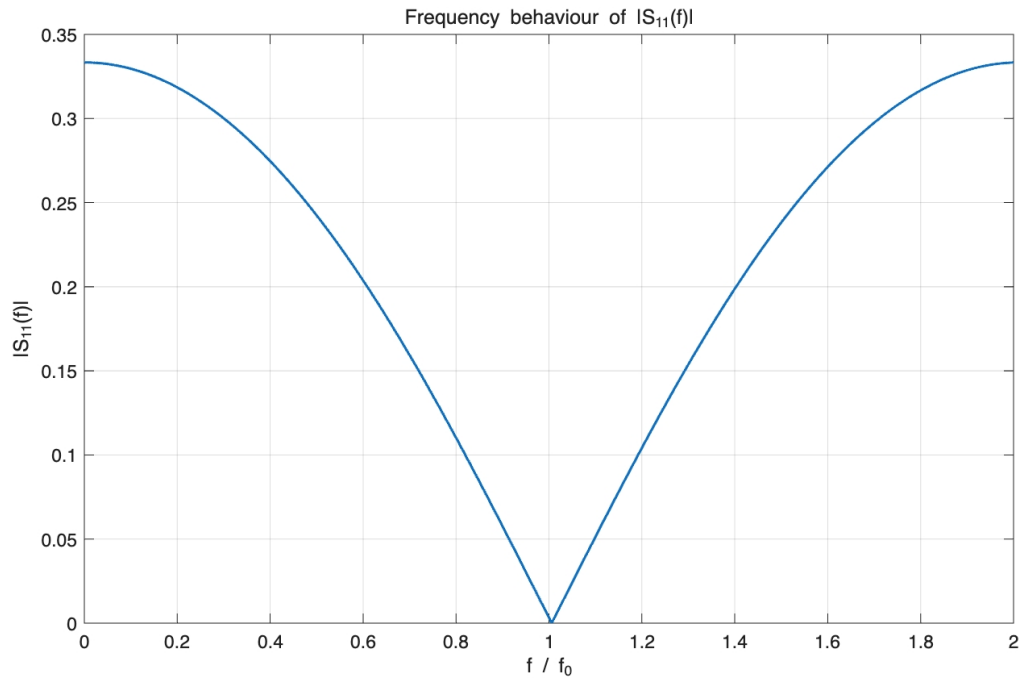
$$Z_{in}(f) = \frac{Z_{in,2}(f) Z_{in,3}(f)}{Z_{in,2}(f) + Z_{in,3}(f)}$$

And

$$S_{11}(f) = \frac{Z_{in}(f) - Z_0}{Z_{in}(f) + Z_0}$$

Thus we can compute $S_{11}(f)$ in Matlab.

We then get a figure as below:



From the figure we can see that the minimum value of $|S_{11}(f)|$ occurs at the design frequency f_0 . At this frequency, the input port is perfectly matched, so there is no reflected wave.

This result also confirms the expected behavior of a QWT: the perfect matching is obtained only at the design frequency.

Problem 2

1. A T-junction without QWT should be selected.

This is because if we choose branch characteristic impedances equal to the corresponding load impedances, which is

$$Z_{\infty 2} = Z_{L2} = 112.5\Omega, \quad Z_{\infty 3} = Z_{L3} = 90\Omega$$

so the input impedances seen from point A are 112.5Ω and 90Ω . Their parallel combination is

$$112.5 \parallel 90 = \frac{112.5 \times 90}{112.5 + 90} = 50\Omega$$

which is equal to the input reference impedance at port 1. Therefore, port 1 is directly matched.

2. As mentioned in Question 1,

$$Z_{\infty 2} = 112.5\Omega, \quad Z_{\infty 3} = 90\Omega$$

The current amplitudes are then

$$|I_2| = \frac{|V_A|}{112.5}, \quad |I_3| = \frac{|V_A|}{90}$$

Therefore,

$$\frac{|I_3|}{|I_2|} = \frac{112.5}{90} = 1.25$$

which satisfies the required amplitude ratio.

Moreover, to satisfy the phase condition (according to the Example 3) the two branch lengths must differ by:

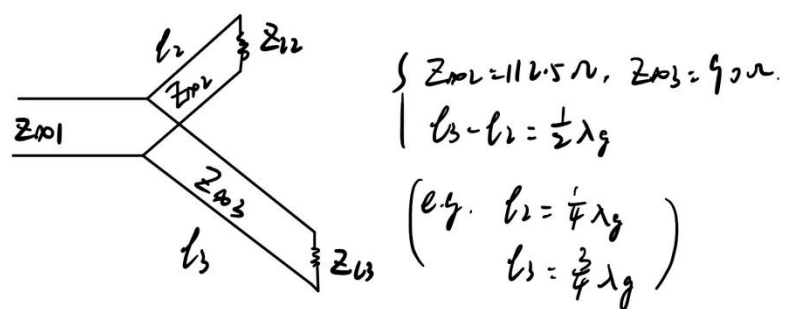
$$\ell_3 - \ell_2 = \frac{\lambda_g}{2}$$

Thus we determine:

$$Z_{\infty 2} = 112.5\Omega, \quad Z_{\infty 3} = 90\Omega$$

$$\ell_2 = \frac{\lambda_g}{4}, \quad \ell_3 = \frac{3\lambda_g}{4}$$

3. Thus we can draw the transmission line equivalent model as below:



4. The load currents is:

$$I_2 = \frac{V_A}{Z_{L2}} e^{-jk\ell_2}, \quad I_3 = \frac{V_A}{Z_{L3}} e^{-jk\ell_3}$$

so

$$\frac{I_3}{I_2} = \frac{Z_{L2}}{Z_{L3}} e^{-jk(\ell_3 - \ell_2)}$$

substituting with the results of Question 2, we obtain:

$$\frac{I_3}{I_2} = 1.25 e^{-jk\lambda_g/2}$$

with

$$k = \frac{2\pi f \sqrt{\epsilon_{\text{eff}}}}{c}, \quad \lambda_g = \frac{c}{f_0 \sqrt{\epsilon_{\text{eff}}}}$$

therefore we get

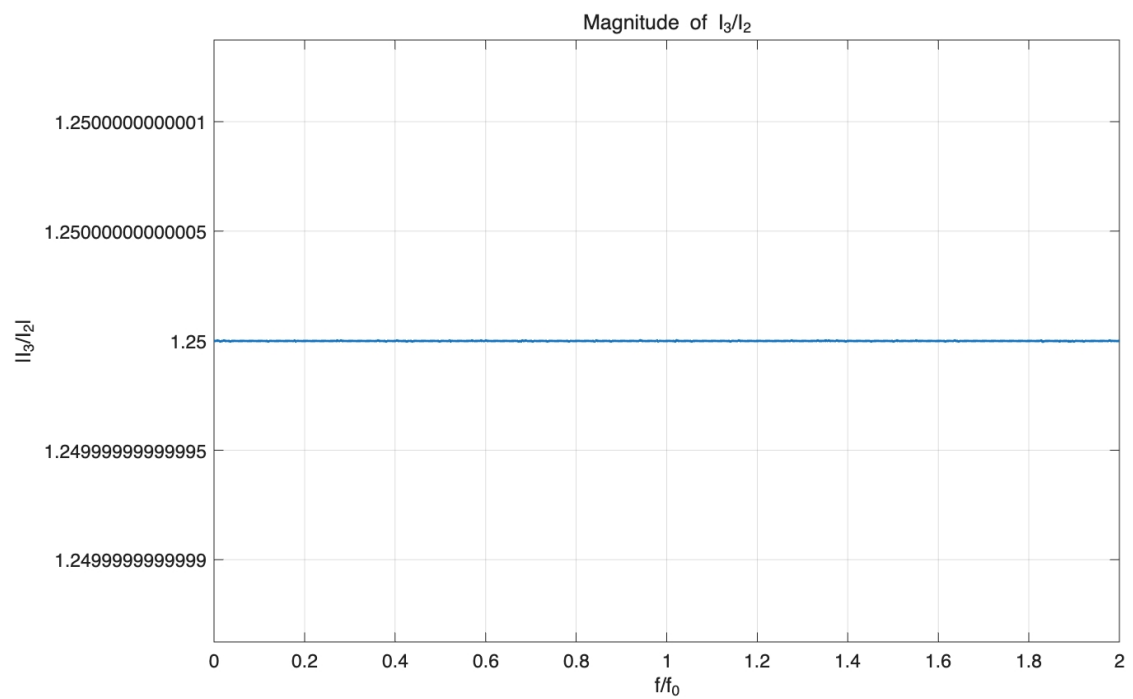
$$\frac{I_3}{I_2} = 1.25 e^{-j\pi(f/f_0)}$$

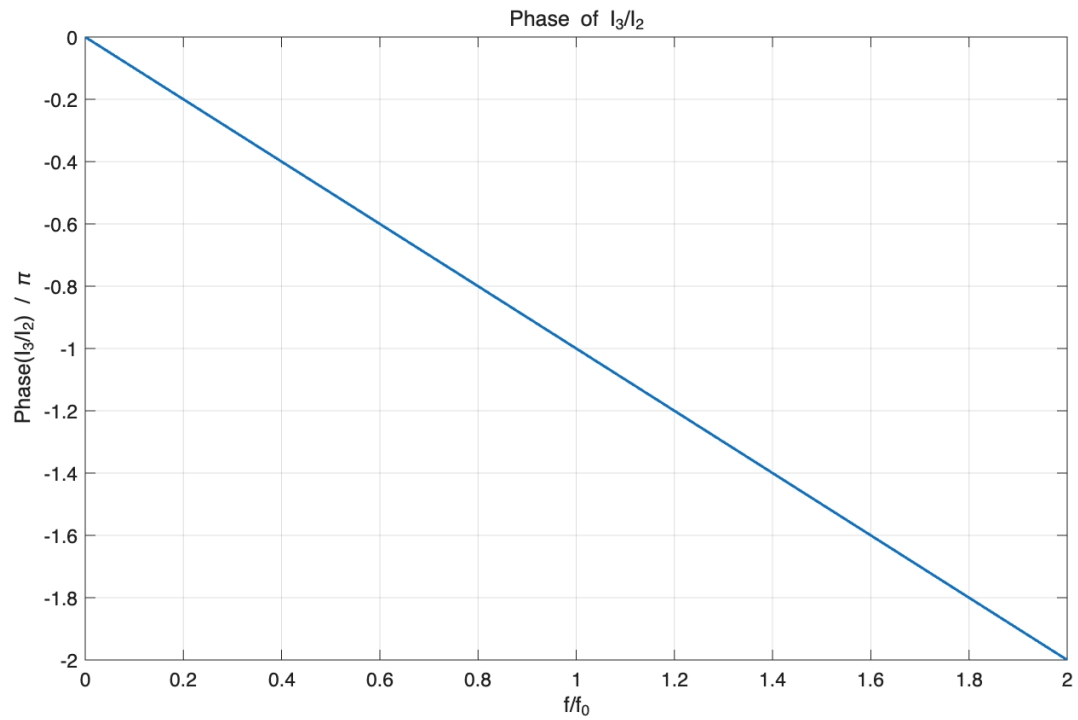
From this equation we can get:

$$\left| \frac{I_3}{I_2} \right| = |1.25 e^{-j\pi(f/f_0)}|$$

$$\angle \left(\frac{I_3}{I_2} \right) = -\pi \frac{f}{f_0}$$

We then plot the figure in matlab:





From the figures we can conclude:

The magnitude is constant and equal to 1.25 over the whole frequency range, which means the amplitude condition is always satisfied.

The phase decreases linearly with $\frac{f}{f_0}$. At $\frac{f}{f_0}=1$, the phase is π , so the required phase condition is satisfied at the design frequency.